

Skin Lesion Analysis

SQL Healthcare Data Analysis

Exploring demographics, lesion characteristics and recorded risk factors

TOOLS

PostgreSQL

SQL • joins • aggregation

FOCUS

Healthcare

descriptive analytics

OUTPUT

Portfolio

analysis • insights • recommendations

OBJECTIVE

Explore

patient demographics, lesion characteristics, lifestyle factors and environmental variables associated with skin lesion diagnoses.

DATA MODEL

2 tables

patient-level characteristics joined to lesion-level diagnostic records using patient_id.

APPROACH

Descriptive

SQL analysis to identify patterns and associations; not causal or predictive modelling.

- Transform coded diagnosis values into analytically useful malignant and benign groupings.
- Compare demographic and behavioural groups using counts and within-group percentages.
- Explore lesion diameter, growth and symptom patterns.
- Translate results into cautious, evidence-aligned recommendations for further analysis.

STEP 1

Join

Connect patient and lesion tables

STEP 2

Transform

CASE classifications and age bands

STEP 3

Aggregate

COUNT, AVG, GROUP BY

STEP 4

Compare

FILTER, percentages and window calculations

STEP 5

Interpret

Separate descriptive findings from causal claims

Portfolio SQL also demonstrates ORDER BY, UNION ALL and conditional aggregation.

Demographic Findings

FINDINGS

AGE GROUP

460

lesion diagnoses among adults aged 60–79 — the largest age-group count.

MALE PATIENTS

66.7%

726 of 1,088 patients in the dataset.

FEMALE PATIENTS

33.3%

362 of 1,088 patients in the dataset.

- Older adults (60–79) accounted for the largest number of lesion diagnoses.
- The dataset contained approximately twice as many male as female patients.
- This describes the sample distribution and does not establish a population-level difference in risk.
- Face, chest, nose, back and forearm recorded the highest counts of malignant lesions.

Lesion Characteristics

FINDINGS

MOST FREQUENT DIAGNOSIS

ACK

Actinic keratosis

BIOPSIED LESIONS

458

lesions recorded as biopsied

LARGEST AVG. DIAMETER

14.09

Melanoma (MEL)

MALIGNANT LESIONS

74.9% growing

259 of 346 malignant lesions were recorded as growing.

BENIGN LESIONS

33.8% growing

251 of 742 benign lesions were recorded as growing.

Growth and diameter are descriptive characteristics in this dataset; they are not presented as standalone clinical predictors.

NO PIPED WATER

782

patients

NO SEWAGE SYSTEM

815

patients

COMMON BIOPSY SITES

Face

then chest, nose, back and forearm

- Face, forearm, chest, back and arm were the most common body sites across diagnosed lesions.
- No clear association was observed between poorer sanitation access and a higher proportion of malignant lesions.
- Within this dataset, malignancy patterns did not appear to differ substantially by sanitation access.

SMOKING

75.8% malignant

47 of 62 lesions among smokers, compared with 29.1% among non-smokers.

ALCOHOL

71.0% malignant

98 of 138 lesions in the alcohol-consumption group, compared with 26.1% without.

PESTICIDE EXPOSURE

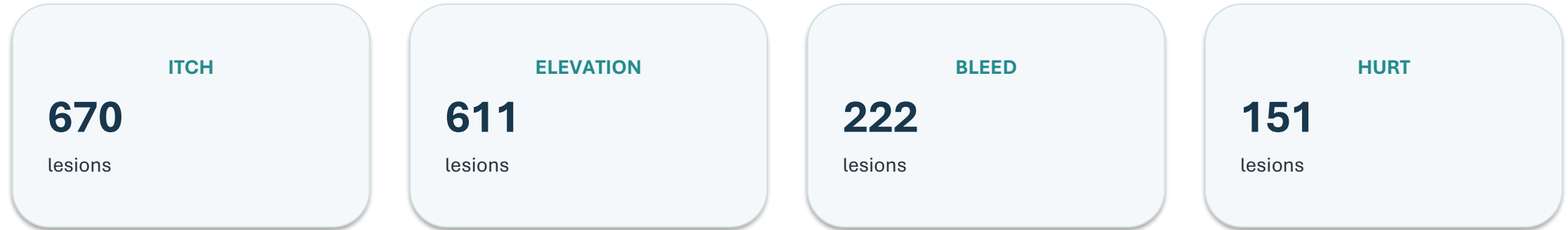
70.0% malignant

156 of 223 lesions with reported exposure, compared with 22.0% without.

- Smoking, alcohol consumption and pesticide exposure were each associated with a higher proportion of malignant lesions in this dataset.
- These comparisons are unadjusted descriptive associations and do not establish causation.

Recorded Symptoms

FINDINGS



- Itch and elevation were the most frequently recorded symptoms.
- Symptom frequency alone does not determine malignancy; further multivariable analysis would be needed to assess independent associations.

- Older adults accounted for the largest number of lesion diagnoses.
- Malignant lesions were more frequently recorded as growing than benign lesions (74.9% vs 33.8%).
- Melanoma had the largest average lesion diameter (14.09).
- Smoking, alcohol consumption and pesticide exposure were associated with higher proportions of malignant lesions.
- No clear association was observed between poorer sanitation access and malignancy.

Findings describe associations within the dataset and do not establish causation or population-level risk.

Recommendations for Further Analysis

NEXT STEPS

VALIDATE

Repeat the analysis using a larger and more representative dataset.

ADJUST

Use multivariable analysis to explore smoking, alcohol and pesticide exposure while accounting for confounding.

COMBINE

Assess lesion growth, diameter and symptoms together rather than as isolated characteristics.

VISUALISE

Develop an interactive dashboard to monitor demographic, diagnostic and lesion trends.

GOVERN

Avoid translating descriptive associations into clinical risk recommendations without further validation.

Portfolio Skills Demonstrated

CLOSE

SQL

PostgreSQL

INNER JOIN • CASE • GROUP BY • FILTER •
UNION ALL

ANALYSIS

Healthcare

descriptive statistics • subgroup comparisons
• interpretation

COMMUNICATION

Evidence-led

clear findings • limitations • next-step
recommendations

Marc Riley | Healthcare Analytics Portfolio

Skin lesion analysis using PostgreSQL

Simulated healthcare analytics scenario • No real patient-identifiable information presented.